Student Name: _____

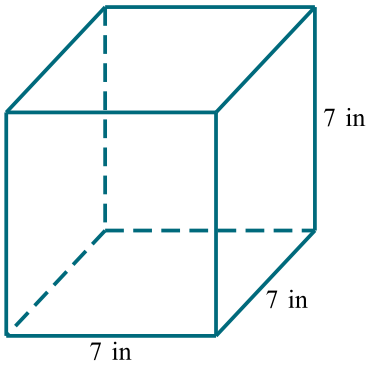
Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **31**

Instructor Name: **Bodanske, Daniel**

Question 1 of 31

Find the volume of the rectangular prism.

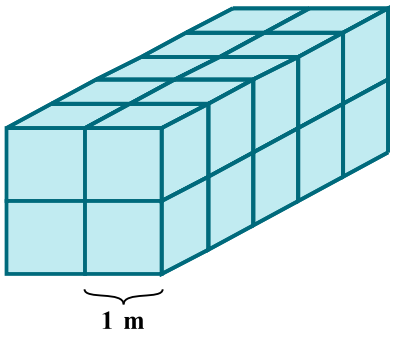


Question 2 of 31

A company makes concrete bricks that are shaped like rectangular prisms. Each brick is 5 inches long, 4 inches wide, and 3 inches tall. How much concrete will they need to make 1500 bricks?

Question 3 of 31

The solid below is made from cubes.
Find its volume.

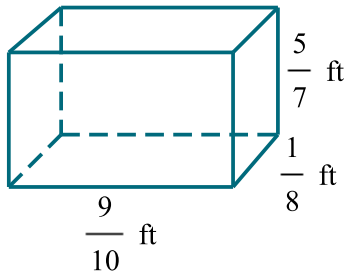


_____ m^3

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

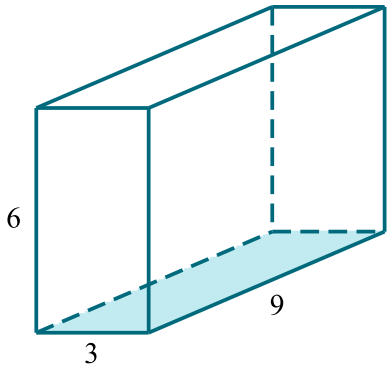
Question 4 of 31

Find the volume of the rectangular prism.
Write your answer in simplest form.



Question 5 of 31

The figure below shows a rectangular prism. One base of the prism is shaded.



(a) Write an expression to find the volume of the prism using its length, width, and height. Then simplify.

Volume of prism: ___×___×___=___ cubic units

(b) Find the area of the shaded base. Then, using the area of the shaded base, write an expression to find the volume of the prism and simplify.

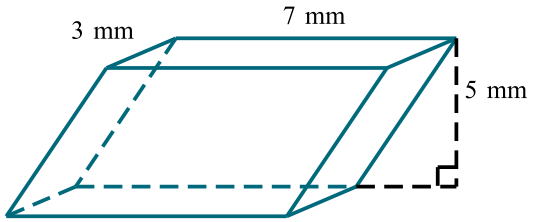
Area of shaded base: ___×___=___ square units

Volume of prism: ___×___=___ cubic units

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 6 of 31

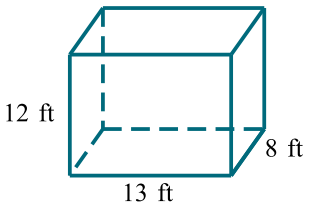
Find the volume of the oblique rectangular prism with length $7\ \text{mm}$, width $3\ \text{mm}$, and height $5\ \text{mm}$.



Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 7 of 31

The figure below shows a shed in the shape of a rectangular prism.
Note that the shed has a floor.



(a) Use the calculator to find the volume and surface area of the shed.
Make sure to include the correct units.

Volume: _____

Surface area: _____

(b) The shed is used for storage.
Which measure would be used to find how much the shed holds?

○ Volume ○ Surface area

(c) The shed is made of metal.
Which measure helped the shed maker know how much metal was needed?

○ Volume ○ Surface area

(a) Use the calculator to find the volume and surface area of the shed.
Make sure to include the correct units.

Volume:

Surface area:

(b) The shed is used for storage.
Which measure would be used to find how much the shed holds?

○ Volume ○ Surface area

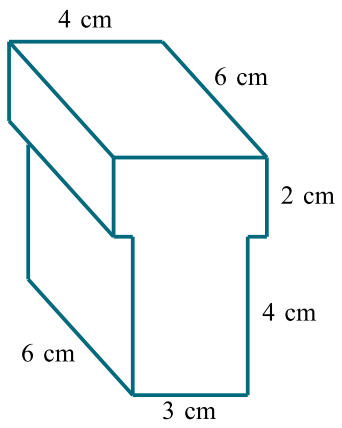
(c) The shed is made of metal.
Which measure helped the shed maker know how much metal was needed?

○ Volume ○ Surface area

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 8 of 31

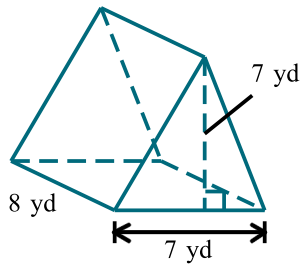
Find the volume of the solid.



Question 9 of 31

Find the volume of this triangular prism.

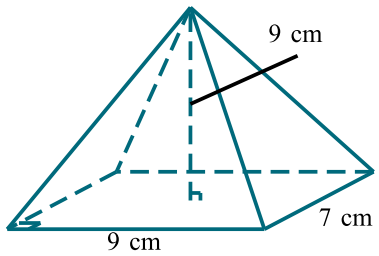
Be sure to include the correct unit in your answer.



Question 10 of 31

Find the volume of this rectangular pyramid.

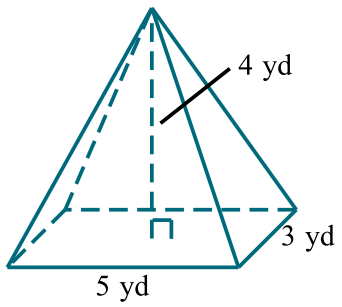
Be sure to include the correct unit in your answer.

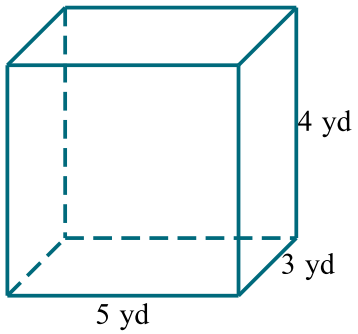


Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 11 of 31

The rectangular pyramid and rectangular prism shown below have the same length, same width, and same height.





Complete the following.

(a) Volume of the pyramid: _____ yd^3

(b) Volume of the prism: _____ yd^3

(c) Volume of the pyramid $=$ _____ $\times$ Volume of the prism

○ This equation is true only for the rectangular prism and rectangular pyramid shown above.

○ This equation is true for all rectangular prisms and rectangular pyramids.

○ This equation is true for all rectangular prisms and rectangular pyramids with the same length, same width, and same height.

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

(a) Find the volume of the pyramid.

_____ yd^3

(b) Find the Volume of the prism.

_____ yd^3

(c) Complete the equation.

Volume of the pyramid = _____ × Volume of the prism

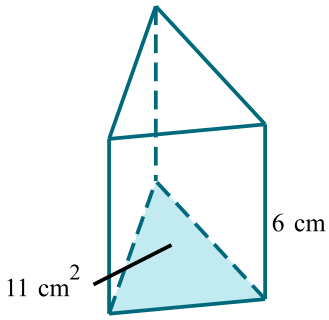
When is this equation true?

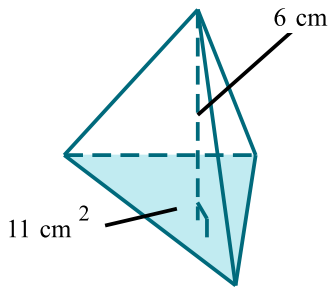
- This equation is true only for the rectangular prism and rectangular pyramid shown above.
- This equation is true for all rectangular prisms and rectangular pyramids.
- This equation is true for all rectangular prisms and rectangular pyramids with the same length, same width, and same height.

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 12 of 31

The triangular prism and triangular pyramid shown below have congruent bases and equal heights.





Complete the following. Write your answers in simplest form.

(a) Volume of the prism: ____ cm^3

(b) Volume of the pyramid: ____ cm^3

(c) Volume of the pyramid = ____ × Volume of the prism

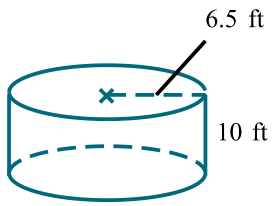
- This equation is true only for the triangular prism and triangular pyramid shown above.
- This equation is true for all triangular prisms and triangular pyramids.
- This equation is true for all triangular prisms and triangular pyramids with congruent bases and equal heights.

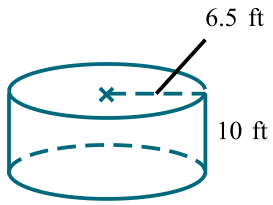
Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 13 of 31

The radius of a cylindrical water tank is 6.5 ft, and its height is 10 ft. What is the volume of the tank?

Use the value 3.14 for π, and round your answer to the nearest whole number.
Be sure to include the correct unit in your answer.





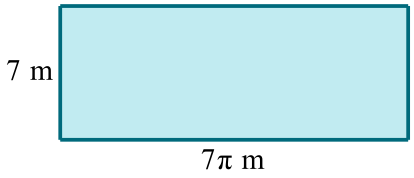
Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

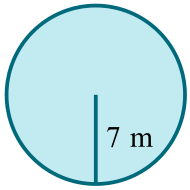
Question 14 of 31

Answer the following questions.
Write your answers in terms of π.

Give exact answers (not decimal approximations).

(a) The rectangle and the circle below have the same area.
What is the area?

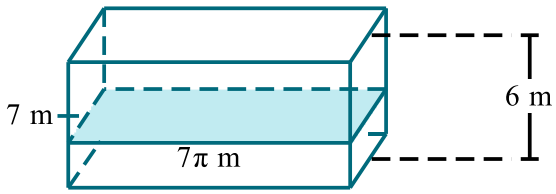




(Radius is 7 m)

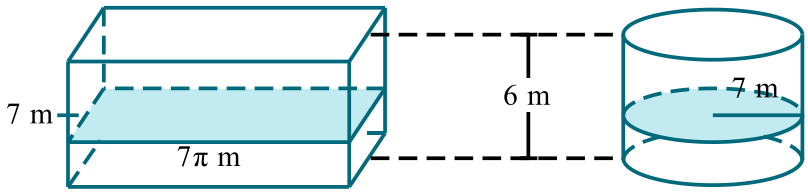
Area of rectangle $=$ _____ m^2 Area of circle $=$ _____ m^2

(b) The rectangle in part (a) is a cross section parallel to the base of the prism below. The height of the prism is 6 m. Find the volume of the prism.



Volume of prism $=$ _____ m^3

(c) The prism and cylinder below have the same height (6 m).
A plane parallel to the bases crosses both solids at the same level.
The resulting cross sections (shaded) are the rectangle and the circle in part (a). Note that they have the same area. Following from Cavalieri's Principle, the prism and the cylinder must have the same volume. Use part (b) to find the volume of the cylinder.



Volume of cylinder $=$ _____ m^3

(d) Let r be the radius and h be the height of the cylinder.
In terms of these variables, which of the following gives the volume of the cylinder in part (c)?

- ○ Volume $= \pi r^2 h$
- ○ Volume $= \dfrac{\pi r^2 h}{3}$
- ○ Volume $= \pi r h$
- ○ Volume $= r^2 h$

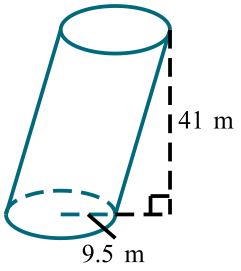
Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 15 of 31

Find the volume of an oblique cylinder with a base radius of $9.5\ \mathrm{m}$ and a height of $41\ \mathrm{m}$.

Use the value 3.14 for π, and round your answer to the nearest whole number.

Be sure to include the correct unit in your answer.

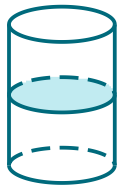
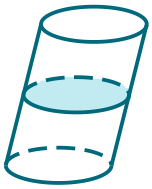
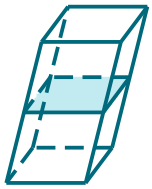


Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

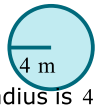
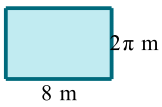
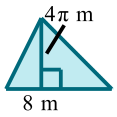
Question 16 of 31

A right cylinder, a slanted cylinder, a slanted rectangular prism, and a right triangular prism are shown below. A plane parallel to the bases crosses the solids at the same level. The resulting cross sections (shaded) are shown.

Answer the following questions. Note that the figures are not drawn to scale.

Solid A	Solid B	Solid C	Solid D
			

(a) Find the areas of the cross sections.
Use the value 3.14 for π. Do not do any rounding.

 (Radius is 4 m)	(Radius is 4 m)		
Area $=$ ___ m^2	Area $=$ ___ m^2	Area $=$ ___ m^2	Area $=$ ___ m^2

(b) If the heights of the solids are the same, which of these solids have the same volume as Solid A? Check all that apply.

- ☐ Solid B
- ☐ Solid C
- ☐ Solid D
- ☐ None of these

(c) Which of the following was most closely used in getting the answer to part (b)?

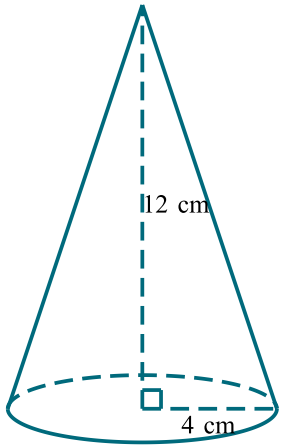
- ○ As long as solids have the same height, they have the same volume.
- ○ As long as solids have the same cross sectional area at a certain level, they have the same volume.
- ○ If solids have the same height and the same cross sectional area at every level, then they have the same volume.
- ○ Even if solids have the same height and the same cross sectional area at every level, they don't necessarily have the same volume.

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 17 of 31

Find the volume of a cone with a base radius of $4\ \text{cm}$ and a height of $12\ \text{cm}$.

Use the value 3.14 for π, and do not do any rounding.
Be sure to include the correct unit in your answer.



Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 18 of 31

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Answer the following questions.
Write your answers in terms of π.

Give exact answers (not decimal approximations).

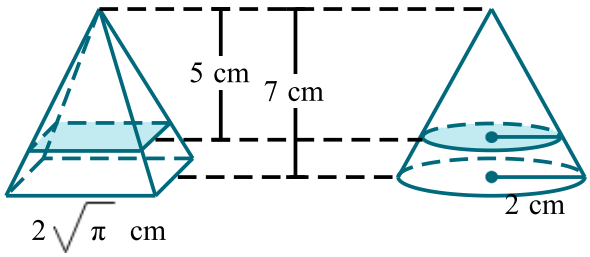
Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

(a) The square pyramid and the cone below both have a height of 7 cm.

The pyramid has a base length of $2\sqrt{\pi}$ cm, and the base of the cone has a radius of 2 cm. A plane parallel to the bases crosses both solids at 5 cm from the top. The resulting cross sections (shaded) have the same area.

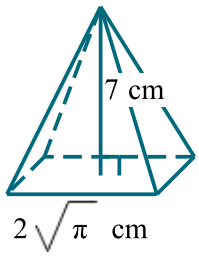
For each solid, the top portion (which has the highlighted cross section as its base) is similar to the entire solid.

Use this fact to find the areas of the cross sections.



Area $=$ _____ cm^2 Area $=$ _____ cm^2

(b) The height of the pyramid is 7 cm. Find the volume of the pyramid.



Volume of pyramid $=$ _____ cm^3

(c) The pyramid and the cone in part (a) have the same height. Their cross sections at every level have the same area. Following from Cavalieri's Principle, the pyramid and the cone must have the same volume.

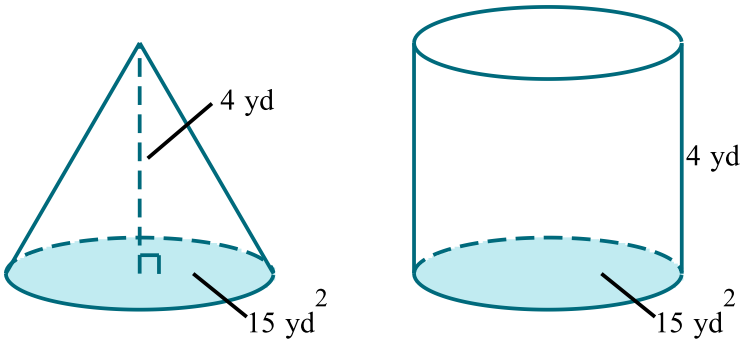
Using these facts, choose the equation that gives the volume in terms of r, the radius of the base of the cone, and h, the height of the cone.

- ○ Volume $=\pi r^2 h$
- ○ Volume $=\dfrac{\pi r^2 h}{3}$
- ○ Volume $=\dfrac{\pi r h}{3}$
- ○ Volume $=2\pi r h$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 19 of 31

The cone and cylinder shown below have congruent bases and equal heights.



Complete the following.

(a) Volume of the cone: ____ yd^3

(b) Volume of the cylinder: ____ yd^3

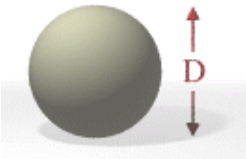
(c) Volume of the cone = ____ × Volume of the cylinder

- This equation is true only for the cylinder and cone shown above.
- This equation is true for all cylinders and cones with congruent bases and equal heights.
- This equation is true for all cylinders and cones.

Question 20 of 31

The diameter, D, of a sphere is 17.6 m. Calculate the sphere's volume, V.

Use the value 3.14 for π, and round your answer to the nearest tenth. (Do not round any intermediate computations.)



Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 21 of 31

A company makes wax candles in the shape of a solid sphere. Suppose each candle has a diameter of $24\ \text{cm}$.

If the company has a total of $180{,}864\ \text{cm}^3$ of wax, how many candles can be made?

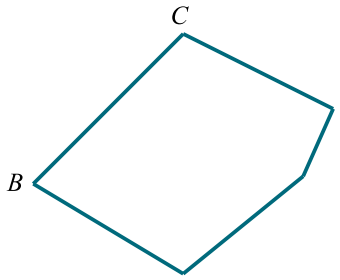
Use 3.14 for π, and do not round your answer.

Question 22 of 31

A pair of similar figures is given below. (Note: The figures are not drawn to scale.)
The sides $\overline{BC}$ and $\overline{DE}$ are corresponding sides.

Use the information below to find BC in Figure 1 and the perimeter of Figure 2.

Figure 1

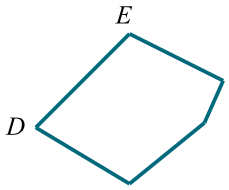


Area of Figure 1: $320\ \text{m}^2$

Perimeter of Figure 1: $40\ \text{m}$

$BC =$ _____ m

Figure 2



Area of Figure 2: $125\ \text{m}^2$

Perimeter of Figure 2: _____ m

$DE = 5\ \text{m}$

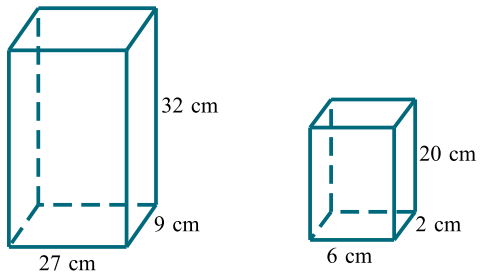
Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 23 of 31

For each pair of solids, determine whether they are similar.
If they are, give the scale factor to get Solid 1 from Solid 2 in simplest form.

(Note: The figures are not drawn to scale.)

(a) The solids are right rectangular prisms.
Their lengths, widths, and heights are shown.



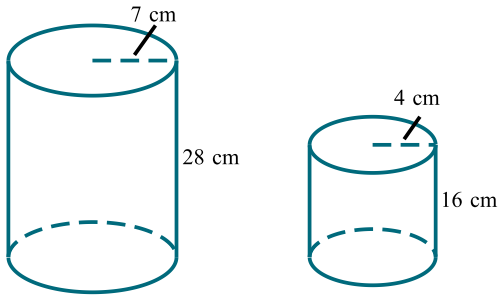
Solid 1 **Solid 2**

○ Similar

Scale factor = _____

○ Not similar

(b) The solids are right cylinders.
Their heights and base radii are shown.



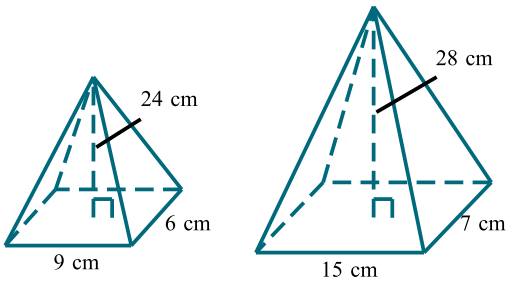
Solid 1 **Solid 2**

○ Similar

Scale factor = _____

○ Not similar

(c) The solids are right rectangular pyramids.
Their lengths, widths, and heights are shown.



Solid 1 **Solid 2**

○ Similar

Scale factor = _____

○ Not similar

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 24 of 31

The ratio of the diameters of two spheres is $5:3$.

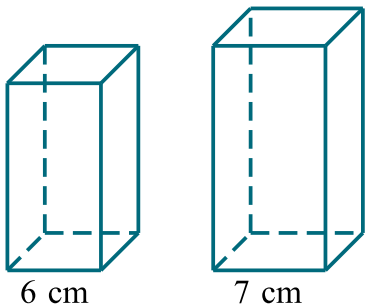
(a) Find the ratio of their volumes.

(b) Find the ratio of their surface areas.

Write these ratios in the format **m:n**.

Question 25 of 31

Two similar prisms have widths as shown below.
(Note: The figures are not drawn to scale.)



(a) If the surface area of the prism on the left is $504\ \text{cm}^2$, find the surface area of the prism on the right.

_____ cm^2

(b) If the volume of the prism on the left is $216\ \text{cm}^3$, find the volume of the prism on the right.

_____ cm^3

(c) If the height of the prism on the left is $12\ \text{cm}$, find the height of the prism on the right.

_____ cm

Question 26 of 31

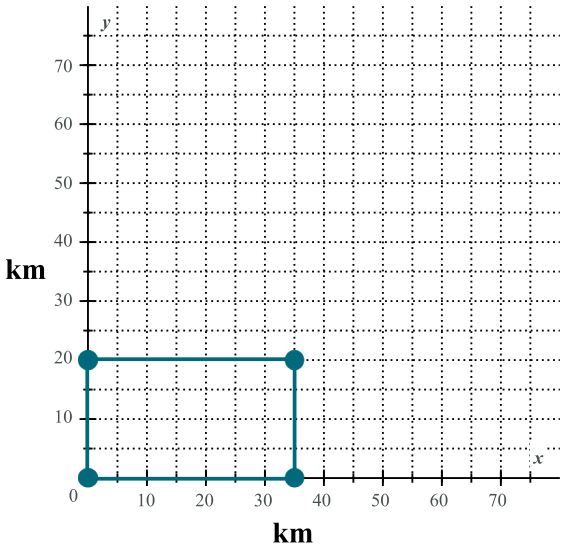
An art collector buys a replica of a famous sculpture. The replica has exactly the same shape as the original but is smaller. The replica is $12\ \text{cm}$ tall, and the original sculpture is $28\ \text{cm}$ tall. If $54\ \text{cm}^3$ of bronze was needed to make the replica, how much bronze was needed to make the original sculpture?

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 27 of 31

A map of a city shaped like a rectangle is given below on a grid. In 2008, there were $1190\ \frac{\text{people}}{\text{km}^2}$ (people per square kilometer) living in the city. What was the population in 2008?

Be sure to include the correct unit in your answer.



Question 28 of 31

Answer the questions below.

Give exact integer or decimal answers.
Be sure to include the correct units in your answers.

(a) An object has a mass of $256\ \text{g}$ and a density of $8\ \frac{\text{g}}{\text{cm}^3}$.

What is the volume of the object?

Volume =

(b) Another object has a volume of $60\ \text{cm}^3$ and a density of $24\ \frac{\text{g}}{\text{cm}^3}$.

What is the mass of the object?

Mass =

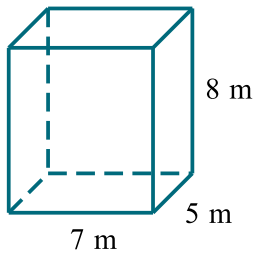
Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 29 of 31

A solid object in the shape of a rectangular prism is shown below.

If its density is $19.6\ \frac{\text{kg}}{\text{m}^3}$, what is its mass?

Give an exact integer or decimal answer.
Be sure to include the correct unit in your answer.

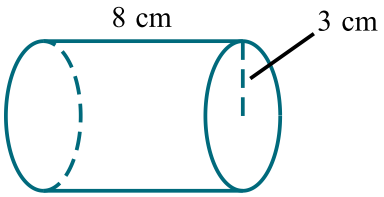


Question 30 of 31

A solid object in the shape of a cylinder is shown below.
It has a radius of $3\ \text{cm}$ and a height of $8\ \text{cm}$.

If its density is $12.7\ \frac{\text{g}}{\text{cm}^3}$, what is its mass?

Use the value 3.14 for π.
Round your answer to the nearest hundredth. Do not round any intermediate steps.
Be sure to include the correct unit in your answer.



Question 31 of 31

Simplify.

$$8\sqrt{5} - 4\sqrt{5}$$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **31**

Question 1 of 31

343 in^3

Question 2 of 31

$90{,}000 \text{ in}^3$

Question 3 of 31

20 m^3

Question 4 of 31

$\frac{9}{112} \text{ ft}^3$

Question 5 of 31

(a) Write an expression to find the volume of the prism using its length, width, and height. Then simplify.

Volume of prism: $9 \times 3 \times 6 = 162$ cubic units

(b) Find the area of the shaded base. Then, using the area of the shaded base, write an expression to find the volume of the prism and simplify.

Area of shaded base: $9 \times 3 = 27$ square units

Volume of prism: $27 \times 6 = 162$ cubic units

Question 6 of 31

105 mm^3

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 7 of 31

(a) Use the calculator to find the volume and surface area of the shed.
Make sure to include the correct units.

Volume: $1248\ \text{ft}^3$

Surface area: $712\ \text{ft}^2$

(b) The shed is used for storage.
Which measure would be used to find how much the shed holds?

- (●) Volume
- ( ) Surface area

(c) The shed is made of metal.
Which measure helped the shed maker know how much metal was needed?

- ( ) Volume
- (●) Surface area

Question 8 of 31

$120\ \text{cm}^3$

Question 9 of 31

$196\ \text{yd}^3$

Question 10 of 31

$189\ \text{cm}^3$

Question 11 of 31

(a) $20\ \text{yd}^3$

(b) $60\ \text{yd}^3$

(c) Volume of the pyramid $= \frac{1}{3} \times$ Volume of the prism

- ( ) This equation is true only for the rectangular prism and rectangular pyramid shown above.
- ( ) This equation is true for all rectangular prisms and rectangular pyramids.
- (●) This equation is true for all rectangular prisms and rectangular pyramids with the same length, same width, and same height.

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 12 of 31

Complete the following. Write your answers in simplest form.

(a) Volume of the prism: $66\ \text{cm}^3$

(b) Volume of the pyramid: $22\ \text{cm}^3$

(c) Volume of the pyramid $= \frac{1}{3} \times$ Volume of the prism

- ○ This equation is true only for the triangular prism and triangular pyramid shown above.
- ○ This equation is true for all triangular prisms and triangular pyramids.
- ◉ This equation is true for all triangular prisms and triangular pyramids with congruent bases and equal heights.

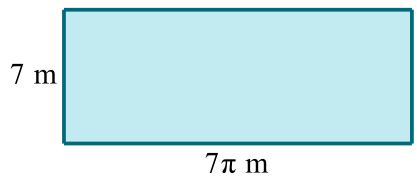
Question 13 of 31

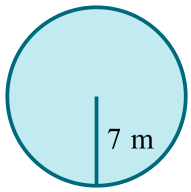
$1327\ \text{ft}^3$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 14 of 31

(a) The rectangle and the circle below have the same area.
What is the area?

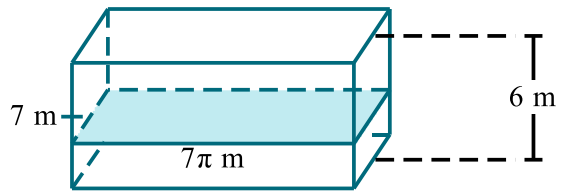




(Radius is $7\ \mathrm{m}$)

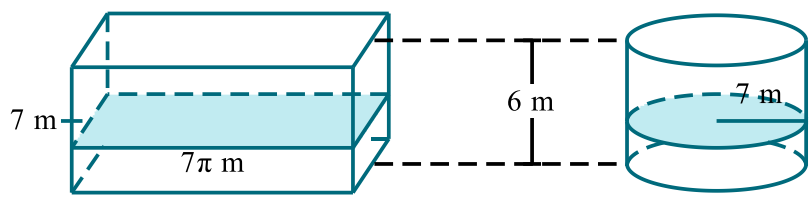
Area of rectangle $= 49\pi\ \mathrm{m}^2$ Area of circle $= 49\pi\ \mathrm{m}^2$

(b) The rectangle in part (a) is a cross section parallel to the base of the prism below. The height of the prism is $6\ \mathrm{m}$. Find the volume of the prism.



Volume of prism $= 294\pi\ \mathrm{m}^3$

(c) The prism and cylinder below have the same height ($6\ \mathrm{m}$).
A plane parallel to the bases crosses both solids at the same level.
The resulting cross sections (shaded) are the rectangle and the circle in part (a). Note that they have the same area. Following from Cavalieri's Principle, the prism and the cylinder must have the same volume. Use part (b) to find the volume of the cylinder.



Volume of cylinder $= 294\pi\ \mathrm{m}^3$

(d) Let r be the radius and h be the height of the cylinder.
In terms of these variables, which of the following gives the volume of the cylinder in part (c)?

- (●) Volume $= \pi r^2 h$
- ( ) Volume $= \dfrac{\pi r^2 h}{3}$
- ( ) Volume $= \pi r h$
- ( ) Volume $= r^2 h$

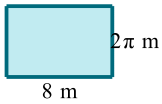
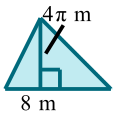
Question 15 of 31

$11{,}619\ \mathrm{m}^3$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 16 of 31

(a) Find the areas of the cross sections.
Use the value 3.14 for π. Do not do any rounding.

(Radius is $4\ \text{m}$)	(Radius is $4\ \text{m}$)		
Area $=50.24\ \text{m}^2$	Area $=50.24\ \text{m}^2$	Area $=50.24\ \text{m}^2$	Area $=50.24\ \text{m}^2$

(b) If the heights of the solids are the same, which of these solids have the same volume as Solid A?
Check all that apply.

- [x] Solid B
- [x] Solid C
- [x] Solid D
- [ ] None of these

(c) Which of the following was most closely used in getting the answer to part (b)?

- ○ As long as solids have the same height, they have the same volume.
- ○ As long as solids have the same cross sectional area at a certain level, they have the same volume.
- ◉ If solids have the same height and the same cross sectional area at every level, then they have the same volume.
- ○ Even if solids have the same height and the same cross sectional area at every level, they don't necessarily have the same volume.

Question 17 of 31

$200.96\ \text{cm}^3$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 18 of 31

(a) The square pyramid and the cone below both have a height of 7 cm.

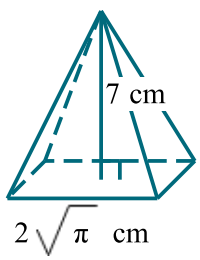
The pyramid has a base length of $2\sqrt{\pi}$ cm, and the base of the cone has a radius of 2 cm. A plane parallel to the bases crosses both solids at 5 cm from the top. The resulting cross sections (shaded) have the same area.

For each solid, the top portion (which has the highlighted cross section as its base) is similar to the entire solid.

Use this fact to find the areas of the cross sections.

Area $= \dfrac{100\pi}{49}$ cm^2 Area $= \dfrac{100\pi}{49}$ cm^2

(b) The height of the pyramid is 7 cm. Find the volume of the pyramid.



Volume of pyramid $= \dfrac{28\pi}{3}$ cm^3

(c) The pyramid and the cone in part (a) have the same height. Their cross sections at every level have the same area. Following from Cavalieri's Principle, the pyramid and the cone must have the same volume.

Using these facts, choose the equation that gives the volume in terms of r, the radius of the base of the cone, and h, the height of the cone.

- ○ Volume $= \pi r^2 h$
- ◉ Volume $= \dfrac{\pi r^2 h}{3}$
- ○ Volume $= \dfrac{\pi r h}{3}$
- ○ Volume $= 2\pi r h$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 19 of 31

Complete the following.

(a) Volume of the cone: $20\ \text{yd}^3$

(b) Volume of the cylinder: $60\ \text{yd}^3$

(c) Volume of the cone $= \frac{1}{3} \times$ Volume of the cylinder

- ○ This equation is true only for the cylinder and cone shown above.
- ◉ This equation is true for all cylinders and cones with congruent bases and equal heights.
- ○ This equation is true for all cylinders and cones.

Question 20 of 31

$V = 2853.1\ \text{m}^3$

Question 21 of 31

25 candles

Question 22 of 31

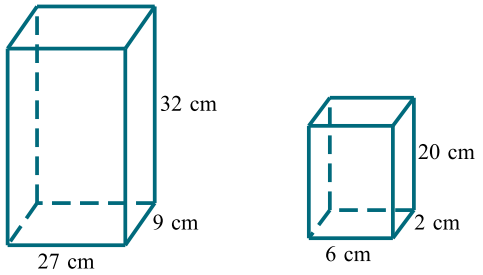
$BC = 8\ \text{m}$

Perimeter of Figure 2: $25\ \text{m}$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 23 of 31

(a) The solids are right rectangular prisms.
Their lengths, widths, and heights are shown.



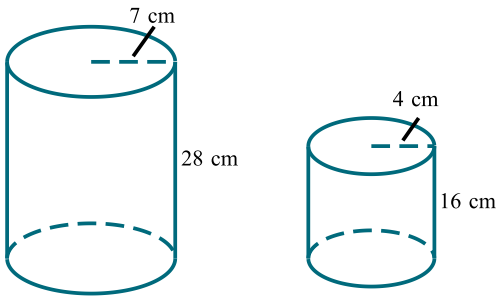
Solid 1 **Solid 2**

- ○ Similar

  Scale factor = _____

- ◉ Not similar

(b) The solids are right cylinders.
Their heights and base radii are shown.



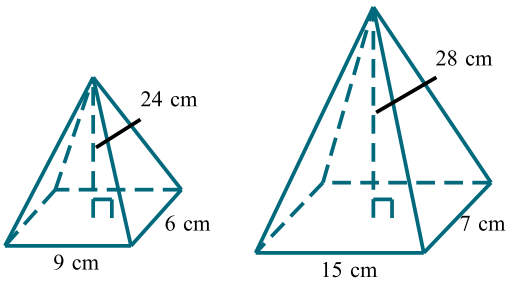
Solid 1 **Solid 2**

- ◉ Similar

  Scale factor = $\frac{7}{4}$

- ○ Not similar

(c) The solids are right rectangular pyramids.
Their lengths, widths, and heights are shown.



Solid 1 **Solid 2**

- ○ Similar

  Scale factor = _____

- ◉ Not similar

Question 24 of 31

(a) $125:27$

(b) $25:9$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 25 of 31

(a) If the surface area of the prism on the left is $504\ \text{cm}^2$, find the surface area of the prism on the right.

$686\ \text{cm}^2$

(b) If the volume of the prism on the left is $216\ \text{cm}^3$, find the volume of the prism on the right.

$343\ \text{cm}^3$

(c) If the height of the prism on the left is $12\ \text{cm}$, find the height of the prism on the right.

$14\ \text{cm}$

Question 26 of 31

$686\ \text{cm}^3$

Question 27 of 31

Population: $833{,}000$ people

Question 28 of 31

(a) An object has a mass of $256\ \text{g}$ and a density of $8\ \frac{\text{g}}{\text{cm}^3}$.

What is the volume of the object?

Volume $= 32\ \text{cm}^3$

(b) Another object has a volume of $60\ \text{cm}^3$ and a density of $24\ \frac{\text{g}}{\text{cm}^3}$.

What is the mass of the object?

Mass $= 1440\ \text{g}$

Question 29 of 31

Mass $= 5488\ \text{kg}$

Question 30 of 31

Mass $= 2871.22\ \text{g}$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation

Question 31 of 31

$4\sqrt{5}$

Copyright © 2026 McGraw Hill - ALEKS® is a registered trademark of ALEKS Corporation